Stable and Efficient value Estimation in Deep Reinforcement Learning

AI Research Paper Explainer

Speed Read

In the research paper, *Deep reinforcement learning with gradient eligibility traces* by Elelimy, Daley, Patterson, Machado, White, and White (2025), the authors introduce a new framework for stable and efficient value estimation in deep reinforcement learning (RL). Popular Deep RL often utilizes semi-gradient descent temporal difference (TD) learning for value estimation. While this has proved to be empirically sufficient as a starting point for some on-policy learning situations, semi-gradient TD is known to diverge under nonlinear function approximation and off-policy learning, which are common in modern RL settings.

In contrast, Gradient TD (GTD) methods provide strong convergence guarantees due to their objectives being well-defined and directly optimized. Despite the theoretical advantages it may have over semi-gradient TD, GTD has historically seen limited adoption in deep RL because early extensions required second-order derivatives and were restricted to one-step updates. More recently, the development of the Generalized Projected Bellman Error (GPBE) allowed GTD to be applied more broadly with nonlinear function approximators, but it remained constrained to single-step learning.

The key contribution of this paper is to extend GPBE into a multistep setting using the λ-return, resulting in the new GPBE(λ) objective. Unlike single-step learning, which provides the agent with the ability to look one step ahead, multi-step learning tries to plan ahead by looking at the sequence of future possible steps and uses that knowledge to make more informed decisions. The λ-return denotes the favorability of a certain action given the weighted average of the one-step return, two-step return, and so on until the N-step return. This extension enables more efficient credit assignment; the assignment of a favorability value to an action by allowing value updates to propagate beyond one step. From this formulation, the authors derive new forward-view and backward-view GTD algorithms, building variants with eligibility traces that are practical for both replay-based and streaming RL.

Besides the theoretical innovations presented in the paper, the authors propose two new deep RL algorithms: Gradient PPO, which integrates GPBE(λ) into Proximal Policy Optimization's critic updates, and QRC(λ), a backward-view control algorithm suited for online streaming tasks. Extensive experiments in MuJoCo continuous control and MinAtar discrete benchmarks show that these new methods outperform their semi-gradient and streaming baselines.

Methodology

The extended GBPE Objective

To understand the changes the authors made in GPBE(λ), we have to look at the original GBPE formula. The Generalized Projected Bellman Error (GPBE), introduced by Patterson, White, and White (JMLR 2022), reformulates the BE in a conjugate optimization form that avoids double sampling, which is a common issue with BE in off-policy deep RL, and works naturally with nonlinear function approximators (like neural networks).

GBPE builds on the original BE formula:

$$BE = \sum_s d(s)E[\delta_t | S_t = s]^2 = \sum_s d(s)\delta_\pi(s)^2$$

Where $d(s)$ is the stationary state distribution, δ_t is the TD error defined as $G_t + \gamma\hat{v}(S_{t+1}) - \hat{v}(S_t)$ with G_t being the immediate return for state S_t and $\hat{v}$ is the parametrized value function for a given state. This means that BE calculates the weighted average expected reward.

$$GBPE = max_{h \in H} \sum_s d(s)[2\delta_\pi(s)\, h(s) - h(s)^2]$$

Building on top of that, GBPE uses the Fenchel conjugate of the quadratic function to make the substitution $\delta_\pi(s)^2 = max_h(2\delta_\pi(s)h - h^2)$, where h is an auxiliary function, and restricts h to a parameterized function class H since we can't optimize all possible functions h.

GPBE(λ) uses the λ-return, which is defined as the exponentially weighted average of n-step returns $G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$, and the associated TD(λ) error $\delta_t^\lambda = G_t^\lambda - \hat{v}(S_t)$. With that objective becomes:

$$GPBE(\lambda) = max_{h \in H} \sum_s d(s)[2\delta_t^\lambda(s)\, h(s) - h(s)^2]$$

When λ is set to 0, the objective reduces to the original GPBE. Intermediate λ balances learning speed, variance, and solution quality.

Proposed Algorithms

From the GPBE(λ), the authors derived three forward-view GTD algorithms under nonlinear function approximation and their backward-view equivalent using eligibility traces: GTD2(λ), TDC(λ), and TDRC(λ). The details of the mentioned algorithms are beyond the scope of this explainer. The forward view TDRC(λ) was used to develop Gradient Proximal Policy Optimization (PPO), a variant of PPO that replaces the squared-error critic objective with a gradient-based loss, and introduces the h function and the λ-returns as shown above. For streaming settings without replay buffers, the authors introduce QRC(λ), an algorithm based on the TDRC(λ) that uses eligibility traces for Q-learning. This algorithm is particularly well-suited for streaming agents where large replay buffers are infeasible (e.g., edge devices or mobile robots).

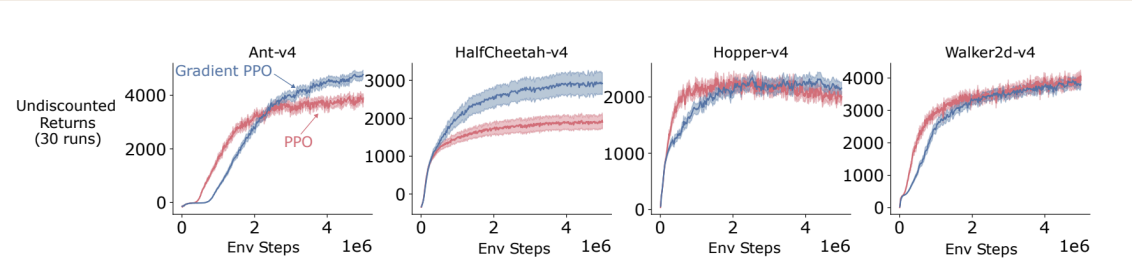
Experimental Setup

- Gradient PPO was evaluated on Ant-v4, HalfCheetah-v4, Hopper-v4, and Walker2d-v4. A hyperparameter sweep determined suitable λ, learning rates, and regularization.
- QRC(λ) was tested against two other Q-learning algorithms, Q(λ) and StreamQ, on five games (Asterix, Breakout, Freeway, Space Invaders, Seaquest), with sweeps over λ and step sizes.
- PPO (with default hyperparameters) and StreamQ (using the authors' implementation) were used as a baseline for evaluation. Each experiment was run over 30 seeds.

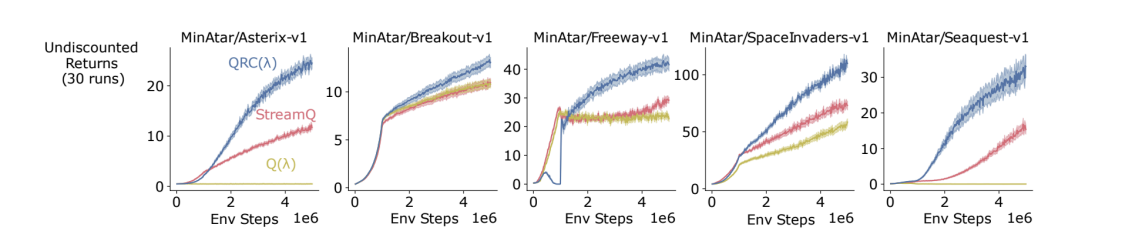
Takeaways

A- Quantitative Results

Gradient PPO significantly outperformed PPO in both Ant and HalfCheetah, and it maintained comparable performance in Hopper and Walker2d.



As for QRC, the algorithm consistently outperformed both Q(λ) and StreamQ across all MinAtar environments. These improvements remained regardless of the usage of a replay buffer.



Overall, the experimental results show a consistent advantage for the use of GBPE(λ), and by extension, λ-returns compared to the baselines and contemporary systems, regardless of the specifics of what agents are used.

Why this Matters at Work

The advancements proposed in this paper open the door to more stable and sample-efficient deep RL agents in both large-scale and resource-constrained settings. One possible use case can be in the field of robotics, where Gradient PPO can be deployed instead of the currently standard PPO, which suffers from instability in the critic that limits its performance. Gradient PPO's multistep GTD critic may enable faster convergence and reduced variance, which is critical in expensive simulation or real-world robotics. Moreover, QRC(λ) provides a practical algorithm for continual learning agents, such as on-device RL for robotics, IoT devices, or adaptive control systems, where replay buffers are infeasible due to memory or privacy constraints. Most importantly, the work presented in this paper advances the broader research community by solidifying the role of GTD methods in deep RL, historically neglected due to perceived inefficiency. GPBE(λ) gives a principled foundation for future algorithm design, potentially influencing actor-critic frameworks beyond PPO.

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